2020 Year 11 Examination

Mathematics Standard

Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	D
3	D
4	A
5	D
6	A
7	B
8	A
9	A
10	D
11	B
12	C
13	B
14	B
15	D

Section II

Question 16

16 (a)

Criteria	Marks
• Correctly expand brackets and simplify to correct answer	2
• One error expanding brackets but correct simplifying afterwards	1

Sample answer

$5(2m-4)-3(2m-5)$
$=10m-20-6m+15$
$=4m-5$

16 (b)

Criteria	Marks
• 2 correct trap rule applications, with summation through to correct answer	3
• 2 correct trap rule applications, with summation using incorrect height	2
• Identifying height = 4	1

Sample answer

$h=4\,, \quad A_1=\frac{4}{2}(0+1.9)=3.8\,,\ A_2=\frac{4}{2}(1.9+3.4)=10.6$
$Paved\ Area \approx 3.8+10.6=14.4m^2$

16 (c)

Criteria	Marks
• Correct calculation and rounding to 2 significant	2
• Correct calculation	1

Sample answer

$r=1.14239\ldots \approx 1.1\ (2\ significant\ figures)$

16 (d)

Criteria	Marks
• Finding the total sales figure	2
• Finding the sales figure, less the first \$10 000	1

Sample answer
$4\% = \$2200$
$100\% = \$55\,000$
$55\,000 + 10\,000 = \$65\,000$

16 (e)

Criteria	Mark
• Identifying sample is 1/5 of the population and multiplying by the correct demographic	1

Sample answer
$\frac{55}{275} = \frac{1}{5}$
$\frac{1}{5} \times 70 = 14 \;\; Year\ 11\ girls$

16 (f)

Criteria	Marks
• Correct time difference (accept both hours and minutes or just minutes)	2
• Adding the longitudes OR calculating 4 minutes per degree	1

Sample answer
$76 + 2 = 78° \qquad 24hrs \div 360° = 4\ minutes/degree$
$78 \times 4 = 312\ mnutes = 5\ hours\ 12\ minutes$

16 (g)

Criteria	Marks
• Converting to correct units and simplifying the rate	2
• Converting 1 hour to 3600 seconds	1

Sample answer
$18000m/3600sec = 5m/sec$

16 (h)

Criteria	Marks
• Correctly converting to calories	2
• Converting fat to kilojoules	1

Sample answer
$6.7 \times 17 = 113.9kJ$
$113.9 \div 4.184 = 27.223\ Calories$

Question 17
17 (a) (i)

Criteria	Mark
• Correct answer	1

Sample answer
$A = 13 \times 6 = 78cm^2$

17 (a) (ii)

Criteria	Marks
• Correctly using Pythagoras to find missing sides then calculate the perimeter	2
• Correctly using Pythagoras theorem to find the missing side	1

Sample answer
$x^2 = 13^2 - 5^2 = 144 \quad x = 12cm$
$P = 6 + 5 + 12 + 6 + 12 + 5 = 46cm$

17 (b)

Criteria	Marks
• Calculating 4 weeks normal pay and adding 17.5% correctly	2
• Calculating 4 weeks normal pay	1

Sample answer
$4weeks = 96\,000 \div 52 \times 4 = 7384.62$
$holiday\ pay = 7384.62 \times 117.5\% = \$8676.92$

17 (c)

Criteria	Marks
• Calculating the yearly cost	3
• Calculating the kW/year value	2
• Calculating the kW/day value	1

Sample answer

$\left((5 \times 11) \div 60\right) \times 9.9 = 9.075kW/day$

$9.075 \times 365 = 3312.375kW/year$

$3312.375 \times 0.2756 = \$912.89$

17 (d) (i)

Criteria	Mark
• Correctly entering values into table	1

Sample answer

	Test indicated LIE	Test did not indicate a LIE	Total
People who LIED	24	6	30
People who did NOT LIE	18	102	120
Total	42	108	150

17 (d) (ii)

Criteria	Marks
• Correctly identifying both cells with accurate results and converting sum into a percentage of the total	2
• Incorrectly identifying both cells with accurate results, and converting sum into a percentage of the total OR converting sum into required percentage	1

Sample answer

$[(LIE - LIE = 24)] + \left[\left(\overline{\overline{LIE}} - \overline{LIE} = 102\right)\right] = 126$

$\frac{126}{150} = 84\%$

17 (e) (i)

Criteria	Mark
• Read off graph	1

Sample answer
$25.

17 (e) (ii)

Criteria	Marks
• Correctly identifying Plan B and identifying the difference as a bonus 50 GB	2
• Correctly identifying Plan B	1

Sample answer
For $50 Plan A would give 350 GB whereas Plan B would give 400 GB. Rohan should use Plan B and get an extra 50 GB.

17 (e) (iii)

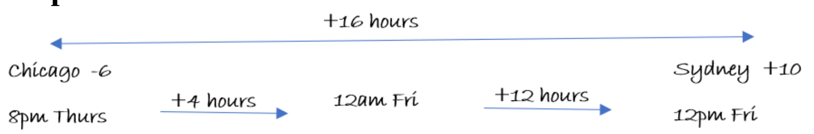
Criteria	Marks
• Correctly calculating the gradient and identifying it as price per Gigabyte	2
• Correctly calculating the gradient	1

Sample answer
$m = \frac{50}{400} = 0.125$
This would be the price per GB, 12.5c per gigabyte.

17 (f)

Criteria	Marks
• Calculating 16 hours difference and using it correctly to find time and day	2
• Calculating 16 hours difference	1

Sample answer



Question 18

18 (a) (i)

Criteria	Mark
• Adding RADIUS of hemisphere to height of cone	1

Sample answer
$150 + 40 = 190cm$

18 (a) (ii)

Criteria	Marks
• Correctly calculating the volume of hemisphere, cone and total	3
• Correctly calculating the volume of hemisphere and cone OR using both formulae correctly with incorrect radius (80cm) and total volume	2
• Correctly calculating the volume of hemisphere or cone OR using both formulae correctly with incorrect radius (80cm)	1

Sample answer
$V_{top} = \frac{1}{2} \times \left(\frac{4}{3} \times \pi \times 40^3\right) = 134\ 041cm^3$
$V_{bottom} = \frac{1}{3} \times \pi \times 40^2 \times 150 = 251\ 327cm^3$
$V_{total} = 13\ 401 + 251327 = 385\ 368cm^3$

18 (b) (i)

Criteria	Mark
• Correctly calculates tax value from table	1

Sample answer
$tax = 20797 + 0.37 \times 8750 = \$24\ 034.50$

18 (b) (ii)

Criteria	Mark
• Correctly finding 2% of taxable income	1

Sample answer

$98750 \times 2\% = \$1975$

18 (b) (iii)

Criteria	Marks
• Sum Tax and Levy then subtracting from tax paid, with correct conclusion	2
• Summing Tax and Levy	1

Sample answer

$24034.50 + 1975 = 26009.50$

$28080 - 26009.50 = 2070.50$

Analise will get a refund of \$2070.50

18 (c)

Criteria	Marks
• Correctly expands the brackets, removes denominator and simplifies correctly to answer	3
• Correctly expands the brackets and removes denominator	2
• Correctly expands the brackets	1

Sample answer

$3x - 30 = 3 - \frac{2x}{3}$

$3x - 33 = -\frac{2x}{3}$

$9x - 99 = -2x$

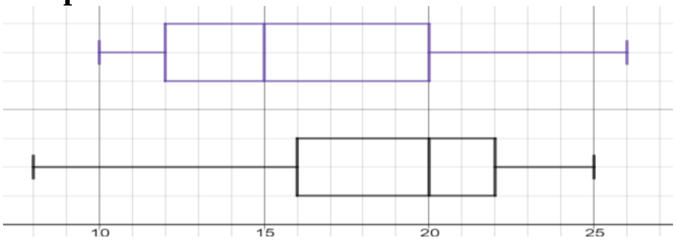
$11x = 99$

$x = 9$

18 (d) (i)

Criteria	Marks
• Correctly drawn graph	2
• Significant progress towards correctly drawn graph (only one error)	1

Sample answer



18 (d) (ii)

Criteria	Marks
• Valid conclusion with justification referring to stats and skewness of plots	3
• Valid conclusion with justification referring to stats OR skewness of plots	2
• Valid conclusion with incorrect justification referring to stats or skewness	1

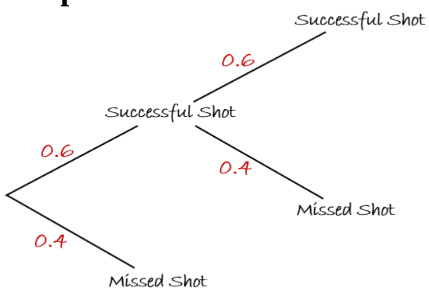
Sample answer
Tim had the better results, with 50% of his scores higher than 20 compared to only 25% of Sara's. Tim's graph has a negative skew showing that his scores are clustered towards the upper end, which is the opposite of Sara's positive skew and clustered results at the lower end.

Question 19

19 (a) (i)

Criteria	Mark
• Correctly labels ALL branches of the probability tree	1

Sample answer



19 (a) (ii)

Criteria	Marks
• Lists all 3 correct outcomes AND their corresponding points	2
• Lists all 3 correct outcomes but not their corresponding points OR has additional incorrect outcomes	1

Sample answer

S=successful shot, M=miss

Outcomes: SS (2 points), SM (1 point), M (0 points)

19 (a) (iii)

Criteria	Mark
• Correctly calculates multistage probability	1

Sample answer

$P(SS) = 0.6 \times 0.6 = 0.36$

19 (a) (iv)

Criteria	Mark
• Correctly identifies the outcome with the highest probability	1

Sample answer
$P(SM) = 0.6 \times 0.4 = 0.24 \ \ and \ \ P(M) = 0.4$
A score of zero has the highest likelihood.

19 (b)

Criteria	Marks
• Correctly makes A the subject of the equation	2
• Correctly makes Ae^{mt} the subject of the equation	1

Sample answer
$P = Ae^{mt} + 1000$

$Ae^{mt} = P - 1000$
$A = \frac{P - 1000}{e^{mt}}$

19 (c) (i)

Criteria	Marks
• Correctly analyses values, IQR, outlier calculation and conclusion	3
• Correctly analyses values and IQR	2
• Correctly analyses values	1

Sample answer
$\bar{x} = 60.4 \quad \sigma = 12.6 \quad Q_1 = 54 \quad Q_3 = 64$
$IQR = 64 - 54 = 10$
$Q_3 + 1.5 \times IQR = 64 + 1.5 \times 10 = 79$
∴ 92 is an outlier

19 (c) (ii)

Criteria	Marks
• Correctly analysis values and explanation of the effect of the outlier	2
• Correctly analysis values	1

Sample answer

$\bar{x} = 56.9 \quad \sigma = 7.2 \quad Q_1 = 51.5 \quad Q_3 = 63$

Removing the outlier has removed the distortion that it caused to the mean and standard deviation, making them a better representation of the scores.

19 (d)

Criteria	Marks
• Correctly simplifies to required expression	3
• Identifies that 50% profit means (selling price) = 1.5 x (cost price) AND simplifies the COST of the order to $\frac{7C}{3}$	2
• Identifies that 50% profit means (selling price) = 1.5 x (cost price) OR simplifies the COST of the order to $\frac{7C}{3}$	1

Sample answer

$$Selling\ price = 1.5 \times \left(C + \frac{4C}{3}\right)$$

$$= \frac{3}{2} \times \left(\frac{3C + 4C}{3}\right)$$

$$= \frac{3}{2} \times \frac{7C}{3}$$

$$= \frac{7C}{2}$$

2020 Year 11 Mathematics Standard Mapping Grid

Section I

Question	Marks	Content	Syllabus Outcomes
1	1	Formula and Equations	MS-A1
2	1	Simple Interest	MS-F1.1
3	1	Formula and Equations	MS-A1
4	1	Classifying and Representing Data	MS-S1.1
5	1	Practicalities of Measurement	MS-M1.1
6	1	Working with Time	MS-M2
7	1	Percentage Calculations	MS-F1.1
8	1	Formula and Equations	MS-A1
9	1	Perimeter, area and volume	MS-M1.2
10	1	Linear Relationships	MS-A2
11	1	Practicalities of Measurement	MS-M1.1
12	1	Formula and Equations	MS-A1
13	1	Data Representation	MS-S1.1
14	1	Depreciation	MS-F1
15	1	Summary Statistics	MS-S1.2

Section II

Question	Marks	Content	Syllabus Outcomes
16 (a)	2	Formula and Equations	MS-A1
16 (b)	3	Trapezoidal Rule	MS-M1.2
16 (c)	2	Formula and Equations	MS-A1
16 (d)	2	Earning and Managing Money	MS-F1.2
16 (e)	1	Classifying and Representing Data	MS-S1.1
16 (f)	2	Working with Time	MS-M2
16 (g)	2	Practicalities of Measurement	MS-M1.1
16 (h)	2	Units of Energy and Mass	MS-M1.3
17 (a) (i)	1	Perimeter, area and volume	MS-M1.2
17 (a) (ii)	2	Perimeter, area and volume	MS-M1.2
17 (b)	2	Earning and Managing Money	MS-F1.2
17 (c)	3	Units of Energy and Mass	MS-M1.3
17 (d) (i)	1	Relative Frequency and Probability	MS-S2
17 (d) (ii)	2	Relative Frequency and Probability	MS-S2
17 (e) (i)	1	Linear Relationships	MS-A2
17 (e) (ii)	2	Linear Relationships	MS-A2
17 (e) (iii)	2	Linear Relationships	MS-A2
17 (f)	2	Working with Time	MS-M2
18 (a) (i)	1	Perimeter, area and volume	MS-M1.2
18 (a) (ii)	3	Perimeter, area and volume	MS-M1.2
18 (b) (i)	1	Earning and Managing Money	MS-F1.2
18 (b) (ii)	1	Earning and Managing Money	MS-F1.2
18 (b) (iii)	2	Earning and Managing Money	MS-F1.2
18 (c)	3	Formula and Equations	MS-A1
18 (d)	2	Summary Statistics	MS-S1.2
18 (d) (ii)	3	Summary Statistics	MS-S1.2
19 (a) (i)	1	Relative Frequency and Probability	MS-S2
19 (a) (ii)	2	Relative Frequency and Probability	MS-S2
19 (a) (iii)	1	Relative Frequency and Probability	MS-S2
19 (a) (iv)	1	Relative Frequency and Probability	MS-S2
19 (b)	2	Formula and Equations	MS-A1
19 (c) (i)	3	Summary Statistics	MS-S1.2
19 (c) (ii)	2	Summary Statistics	MS-S1.2
19 (d)	3	Earning and Managing Money	MS-F1.2